INCH-POUND

MIL-C-60945B (AR)
12 May 1993

SUPERSEDING
MIL-C-60945A (AR)
15 September 1980

MILITARY SPECIFICATION

CARTRIDGE, 40MM, PRACTICE, M385A1
DUMMY CARTRIDGE, 40MM HIGH VELOCITY, M922
PROJECTILE PARTS FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirement, examinations and tests for two types of cartridges designated as Cartridge, 40MM, practice, M385A1 and Dummy Cartridge, 40MM, High Velocity, M922 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC 1310**

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SPECIFICATIONS

MILITARY

MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Table for Inspection by Variables for Percent Defective
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings; and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8866714	- Projectile Assembly, For Cartridge, 40MM, Practice, M385A1
12556066	- projectile Assembly for Dummy Cartridge 40MM, High Velocity, M922

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.1.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117 73 - Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing (Dwg) 8866714 or 12556066, all associated drawings and with all requirements specified in applicable specifications.

3.3 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.4 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All require markings and

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stamping shall be neat and sharply defined. Total surface imperfections for nicks, dents, and scratches shall not exceed 2.5% of the surface area of the part. Any individual indentation shall not exceed .010 depth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. (see 3.3)

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Body	8866715	25
Projectile Assembly	8866714 or 12556066	25

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Projectile Assembly and Components	SHEET 1 OF 1	DRAWING NUMBER NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH / INSPECTION METHOD REFERENCE
	Body (Dwg. 8866715)		
	Examination for defects	25	4.4.2.1
	Salt Spray	5	4.5.1
	Projectile Assembly (Dwg. 8866714 or 12556066)		
	Examination for defects	25	4.4.2.2
	Push test of rotating band	10	4.5.2

NOTES:

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4.4. Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examinations and tests. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body	1 OF 2	8866715

NEXT HIGHER ASSEMBLY: 8866714 or 12556066

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined.			
MAJOR				
101.	Diameter of rotating band surface (after knurl)	0.40%	3.2	Gage
102.	Location of basic diameter (excluding cut-off nib)	0.40%	3.2	Gage
103.	Runout of indicated area on taper with body diameter	0.40%	3.2	Gage
104.	Body diameter	0.40%	3.2	Gage
105.	Diameter between knurl	0.40%	3.2	Gage
106.	Width of crest of knurl	0.40%	3.2	Gage
107.	Width of rotating band surface	0.40%	3.2	Gage
108.	Length from back of rotating band surface to front end of node	0.40%	3.2	Gage
109.	Spherical radius	0.40%	3.2	Gage
110.	Salt spray	*	3.2	4.5.1

NOTES:

* Sampling and rejection shall be in accordance with MIL-A-8625.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-60945B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body	2 OF 2	8866715
			NEXT HIGHER ASSEMBLY: 8866714 or 12556066

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
MINOR				
201.	Protective coating with bare spot on bottom or forward of rotating band groove	0.65%	3.2	Visual
202.	Incorrect color	0.65%	3.2	Visual
203.	Evidence of poor workmanship	0.65%	3.4	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body	1 OF 1	8866714 or 12556066
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined.			
MAJOR				
101.	Location of bourrelet shoulder	0.40%	3.2	Gage
102.	Diameter of bourrelet	0.40%	3.2	Gage
103.	Profile of rotating bands	0.40%	3.2	Gage
104.	Diameter of rotating bands	0.40%	3.2	Gage
105.	Diameter of large groove	0.40%	3.2	Gage
106.	Diameter of small groove	0.40%	3.2	Gage
107.	Diameter below bourrelet	0.40%	3.2	Gage
108.	Distance from nose to back of rotating band	0.40%	3.2	Gage
109.	Concentricity of rotating bands with bourrelet and body diameter	0.40%	3.2	Gage
110.	Total length (excluding cut-off nib)	0.40%	3.2	Gage
MINOR				
201.	Marking missing	0.65%	3.2	Visual
202.	Evidence of poor workmanship	0.65%	3.4	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Push test of rotating band on projecting assembly. Beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have been accepted, a sample of fifty (50) assemblies in accordance with MIL-STD-414, Table B-I, with an AQL of 0.25 shall be selected from each band setting machine, each lot, and subjected to this test.

NOTE: Movement as specified on Dwg. 8866714 or 12556066 is defined as movement of the back face of the rotating band relative to the body.

4.4.3.2 Push test to rotating band on projectile assembly. After three (3) acceptable lots have met the criteria of 4.3.3.1, the Government inspector shall select twenty-five (25) assemblies in accordance with MIL-STD-414, Table B-I with an AQL of 0.25 from each band setting machine, each lot and subjected to the test. If any lot fails the above criteria the lots thereafter shall revert to the criteria of 4.3.3.1.

NOTE: Movement as specified on Dwg. 86714 or 12556066 is defined as movement of the back face of the rotating band relative to the body.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Salt spray. This test shall be conducted in accordance with ASTM B 117-73 for 24 hours using 5% salt solution. This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.5.2 Push test of rotating band on projectile assembly. The projectile assembly shall be placed in an approved fixture and the force specified on the drawing shall be applied. Observation shall be made for the requirement specified on the drawing. This test is a destructive test. Parts so tested shall not be returned to the lot.

NOTE: Deformation of the chamfered leading edge of the rotating band caused by compression in the test fixture is permissible.

5. PACKAGING

5.1 Preservation and packing.

5.1.1 Level B. Preservation and packaging shall be in accordance with MIL-STD-1169.

5.2 Packing.

5.2.1 Level B. The projectile assembly preserved and packaged as specified in 5.1.1, shall be packed in accordance with MIL-STD-1169.

5.3 Marking. Marking shall comply with MIL-STD-1169.

5.4 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification is intended for use on Cartridge, 40MM, Practice, M385A1 and Dummy Cartridge, 40MM, High Velocity, M922.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

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c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of results of contractor - conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARDEC, ATTN: SMCAR-QAF-S, Picatinny Arsenal, NJ 07806-5000.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Changes from previous issue. Asterisk are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

M16A2 Metallic Link
M169 Cartridge Case
MK19 Mod 3 Grenade Machine Gun

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A521)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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[redacted]	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-60945B (AR)	930512

3. DOCUMENT TITLE
CARTRIDGE, 40MM, PRACTICE, M385A1 DUMMY CARTRIDGE, 40MM HIGH VELOCITY. M922

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

[redacted]

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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